

Computer Networks & Routing Protocols

A Comprehensive Technical Reference

This document covers the fundamental concepts of computer networking, with a focus on routing protocols, graph algorithms, and network architectures. It serves as a reference for understanding how modern internet infrastructure operates.

Chapter 1: Introduction to Computer Networks

1.1 What is a Computer Network?

A computer network is a set of interconnected devices that communicate with each other using standardized protocols. The Internet is the largest computer network in the world, connecting billions of devices across every continent. Vint Cerf and Bob Kahn designed the TCP/IP protocol suite, which became the foundation of the modern Internet. The OSI model, developed by the International Organization for Standardization (ISO), defines seven layers of network communication: Physical, Data Link, Network, Transport, Session, Presentation, and Application.

1.2 Network Topologies

Network topology refers to the arrangement of nodes and links in a network. A bus topology connects all devices to a single cable. A ring topology forms a circular chain of devices. A star topology uses a central hub or switch to connect all nodes. A mesh topology connects every device to every other device, providing maximum redundancy. The Internet uses a hybrid topology that combines elements of all these types.

1.3 The TCP/IP Protocol Suite

The TCP/IP model uses four layers: the Network Access layer handles physical transmission, the Internet layer uses the IP protocol to route packets, the Transport layer uses TCP for reliable delivery and UDP for fast, connectionless communication, and the Application layer includes protocols like HTTP, DNS, DHCP, and FTP. TCP uses a three-way handshake to establish a connection. UDP does not establish a connection before transmitting data. The IP protocol assigns a unique address to each device on the network.

Chapter 2: Routing and Routing Protocols

2.1 What is Routing?

Routing is the process of selecting a path across one or more networks to send data from a source to a destination. A router is a network device that forwards data packets between

computer networks. Routers use routing tables to determine the best path for each packet. Static routing uses manually configured routes that do not change. Dynamic routing uses protocols that automatically discover and update routes based on the current network topology.

2.2 OSPF — Open Shortest Path First

OSPF is a link-state routing protocol that uses Dijkstra's algorithm to compute the shortest path tree. OSPF was developed by the IETF and is defined in RFC 2328. OSPF uses Hello packets to discover and maintain neighbor relationships. Routers that share a common network segment form an adjacency. OSPF floods Link State Advertisements (LSAs) throughout the network to ensure every router has an identical copy of the link-state database. OSPF divides a network into areas to reduce the volume of routing traffic. Area 0, also called the backbone area, connects all other areas. When a link failure occurs, OSPF detects the failure through the expiration of the Dead Interval timer. The router then floods a new LSA and every router re-runs Dijkstra's algorithm to recalculate the shortest path tree. OSPF supports VLSM and CIDR and is considered a classless routing protocol.

2.3 BGP — Border Gateway Protocol

BGP is the routing protocol that powers the Internet. BGP is a path-vector routing protocol that makes routing decisions based on paths, network policies, and rule-sets. BGP uses TCP port 179 to establish sessions between routers. An Autonomous System (AS) is a collection of IP networks under the control of a single organization. BGP routers exchange Network Layer Reachability Information (NLRI) to advertise reachable prefixes. iBGP runs within the same AS, while eBGP runs between different ASes. The BGP route selection process uses the AS_PATH attribute to detect and prevent routing loops. The LOCAL_PREF attribute determines the preferred exit point from an AS. BGP uses the MED attribute to influence how traffic enters an AS from an external peer.

2.4 RIP — Routing Information Protocol

RIP is one of the oldest distance-vector routing protocols. RIP uses hop count as its metric, with a maximum hop count of 15. A hop count of 16 represents an unreachable network. RIP routers broadcast their entire routing table every 30 seconds. RIPv2 added support for classless routing and multicast updates. RIP uses the Bellman-Ford algorithm to calculate the best path. RIP suffers from the count-to-infinity problem, which is mitigated by split horizon and route poisoning techniques.

2.5 EIGRP — Enhanced Interior Gateway Routing Protocol

EIGRP is an advanced distance-vector routing protocol developed by Cisco. EIGRP uses the Diffusing Update Algorithm (DUAL) to guarantee loop-free paths. EIGRP uses a composite metric based on bandwidth and delay. EIGRP maintains a topology table that stores all routes learned from neighbors. The successor is the primary route to a destination. The feasible successor is a backup route that satisfies the feasibility condition. EIGRP sends partial updates

only when the network topology changes, unlike RIP which sends full updates periodically.

Chapter 3: Graph Algorithms in Networking

3.1 Dijkstra's Algorithm

Dijkstra's algorithm, invented by Edsger Dijkstra in 1956, computes the shortest path from a single source to all other vertices in a weighted graph with non-negative edge weights. OSPF uses Dijkstra's algorithm to build the shortest path tree from each router's perspective. The algorithm maintains a priority queue of unvisited vertices sorted by their current shortest distance. The time complexity of Dijkstra's algorithm is $O((V + E) \log V)$ when implemented with a binary heap. The algorithm produces the shortest path tree (SPT), which OSPF stores in its routing table.

3.2 Bellman-Ford Algorithm

The Bellman-Ford algorithm computes the shortest path from a single source to all other vertices in a weighted graph. Unlike Dijkstra's algorithm, Bellman-Ford can handle negative edge weights. RIP uses the Bellman-Ford algorithm for its distance-vector computations. Bellman-Ford has a time complexity of $O(V * E)$. The algorithm can detect negative weight cycles, which indicate a routing loop in network terms. BGP prevents routing loops using the AS_PATH attribute rather than relying on Bellman-Ford.

3.3 Spanning Tree Protocol

The Spanning Tree Protocol (STP), defined in IEEE 802.1D, prevents Layer 2 loops in Ethernet networks. STP uses a distributed algorithm to elect a root bridge. All switches elect the bridge with the lowest Bridge ID as the root bridge. STP then constructs a spanning tree rooted at the root bridge, disabling redundant links. Rapid Spanning Tree Protocol (RSTP), defined in IEEE 802.1w, converges much faster than STP. Multiple Spanning Tree Protocol (MSTP), defined in IEEE 802.1s, allows multiple VLANs to be mapped to different spanning tree instances.

Chapter 4: Network Security

4.1 Firewalls and Access Control Lists

A firewall is a network security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Cisco developed the first commercial firewall in 1992. A stateful firewall tracks the state of active connections. A stateless firewall filters packets based on static rules without tracking connection state. Access Control Lists (ACLs) are used by routers and switches to filter traffic. An ACL uses permit and deny statements to control which traffic is allowed to pass.

4.2 VPNs and Encryption

A Virtual Private Network (VPN) creates an encrypted tunnel over a public network. IPsec is a protocol suite that provides authentication and encryption at the IP layer. SSL/TLS VPNs use the Transport Layer Security protocol to secure communications. MPLS VPNs use Multi-Protocol Label Switching to create private network segments. WireGuard is a modern VPN protocol that uses state-of-the-art cryptography. OpenVPN is an open-source VPN solution that uses OpenSSL for encryption.

4.3 DNS Security

The Domain Name System (DNS) translates human-readable domain names to IP addresses. DNS uses a hierarchical distributed database. The root nameservers are the top of the DNS hierarchy. DNSSEC adds cryptographic signatures to DNS records to prevent spoofing. DNS over HTTPS (DoH) encrypts DNS queries to protect user privacy. DNS amplification attacks exploit open DNS resolvers to perform distributed denial-of-service attacks.

Chapter 5: Software-Defined Networking (SDN)

5.1 SDN Architecture

Software-Defined Networking (SDN) separates the control plane from the data plane in network devices. The SDN controller is a centralized software application that manages the entire network. OpenFlow is the most widely used protocol for communication between the SDN controller and network devices. Nick McKeown and Scott Shenker pioneered the concept of SDN at Stanford University. The data plane handles packet forwarding based on rules installed by the control plane. SDN enables network programmability and automation through northbound APIs.

5.2 Network Function Virtualization

Network Function Virtualization (NFV) replaces dedicated hardware network appliances with software running on commodity servers. NFV uses virtual machines or containers to host network functions such as firewalls, load balancers, and routers. ETSI defined the NFV

architecture framework. Kubernetes is commonly used to orchestrate containerized network functions. Docker containers provide lightweight isolation for virtualized network functions. NFV and SDN are complementary technologies that together enable cloud-native networking.

Summary: Routing Protocol Comparison

Protocol	Type	Algorithm	Metric	Convergence
OSPF	Link-State	Dijkstra	Cost	Fast
BGP	Path-Vector	Policy-based	AS Path / MED	Slow
RIP	Distance-Vec	Bellman-Ford	Hop Count	Very Slow
EIGRP	Adv. D-Vec	DUAL	BW + Delay	Fast
IS-IS	Link-State	Dijkstra	Cost	Fast

Table 1: Comparison of major IP routing protocols used in the Internet.